



BEST
BAJAJ
ENGINEERING
SKILLS
TRAINING



SASTRA

ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION

DEEMED TO BE UNIVERSITY

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THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

Laboratory Manual

Module No : 1

Module Name : Mechatronics

Sub Module Name: Electrical and Electronics Measurements
using Sensors

Offered by

Bajaj Engineering Skills Training Centre

SASTRA Deemed to be University

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Educational Learning outcomes

This course enables the students to

- Identify and describe the operation of various digital sensors
- Identify and describe the operation of various analog sensors
- Integrate the analog and digital sensors with the PLC units

Industrial Learning Outcomes

This course benefits the students to

- Understand how automation is carried out in continuous process using sensors

Safety Instructions

Note: Wherever this symbol  is shown, it indicates **Caution!** and/or **Warning!**

It indicates that operator error can lead to damage and malfunctioning of the pneumatic equipment and can lead to serious personal injury or loss of life.

Electrical Safety

Following Safety instructions should be strictly followed. Failure to do so may result in accidents, equipment malfunctioning, serious personal injury and/or loss of life.

Electricity is a source of considerable energy. When handling products dealing with electricity, the following precautions must be taken to prevent accidents.

1. Always work on the electrical connections only in the absence of power.
2. Before connecting any electrical equipment to the electrical power supply, all electrical connections should be checked for security. Do not connect the circuit with power until the circuit is finished and checked.
3. No circuit alterations, removal of components, repairing of circuit etc. should be attempted with power supplies connected. Electrical supplies must be disconnected before beginning any adjustment, maintenance or dismantling of equipment.
4. The maximum allowable operating voltage, current, temperature etc. must be strictly observed. Failure to do so might result in catastrophic failure of equipment, and result in serious personal injury and/or death.

Electrical Safety for Electrical Equipment:

Warning

1. Repair & Maintenance

Assembly, handling, or repair of sensor (electrical) systems should be performed by only trained and experienced operators.

2. Safety First

Do not service machinery/equipment or attempt to remove any component until safety is confirmed.

- Inspection and maintenance of machinery/equipment should only be performed after confirmation that electrical supply have been positively disconnected.

3. Contact the concerned in-charge, if equipment is to be used in any of the following conditions:

1. Equipment is to be used in conditions beyond the given specifications, or if equipment is to be used outdoors.
2. Equipment is to be used in conjunction with atomic energy, railroad, air navigation, automobiles or related vehicles, medical equipment or safety equipment.
3. In applications that adversely effect humans, animals, or property requiring special safety analysis.

Installation/ Operating/ Maintenance Instructions

Electrical

1. Sensors

1. Sensors must be mounted within the given mounting provisions and adjustments.
2. Ensure the sensing surface & lens of the sensor are away from interference of any objects or particles that could damage.
3. Contact with chemical substances and hot objects must be prevented

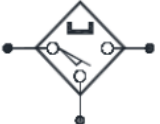
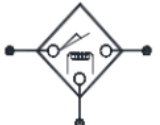
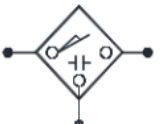
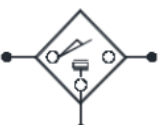
2. Fiber optic cables

1. Fiber optics must not be twisted or kinked.
 2. Excessive tensile loads will lead to destruction of cable.
 3. Contact with petrol or organic solvents must be prevented.
-
3. Note the direction of current flow (i.e. input/output terminals) from the circuit diagram before installation and connect sensor with cable accordingly.
 4. Install in clean atmosphere.
 5. Ensure the cable connection with sensor - until the positive stop
 6. Always ensure your cable is connected as per your required circuit.
 7. Do not switch on the power supply until you have established and secured all electrical connections.
 8. Do not exceed the maximum permissible voltage of 24V DC
 9. Do not disconnect electrical cables under power supply.
 10. Always Safety carries more value than anything. So, handle the instruments as per specification and guidance.
 11. No manual intervention is to take place unless the circuit is at rest.

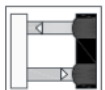
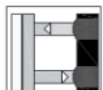
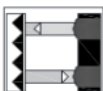
Electrical

1. Electrical connections are to be wired-up or disconnected only when the power supply is turned off
2. Use 24V DC regulated power supply for all the power connections (e.g. SMPS)
3. Ensure that the field devices are being connected appropriately
4. Refer product data sheet for specification and wire connection details for proper usage
5. While connecting sensors. Ensure that the sensor is being connected as follows
 - a) Proximity and Optical sensor
 - a. Brown Positive
 - b. Blue Negative
 - C. Black Output
 - b) Fiber optics sensor:
 - a. Brown Positive
 - b. Blue Negative
 - C. White Teach input
 - d. Black Output
 - c) Colour sensor:
 - a. Brown Positive
 - b. Blue Negative
 - C. Black Output
 - d. Grey Teach input
 - d) Ultrasonic sensor:
 - a. Brown Positive
 - b. Blue Negative
 - C. White Teach input
 - d. Black Analog output

Electrical Symbols

Description	Symbol	Description	Symbol
Magnetic sensor		Relay coil	
Inductive sensor		Normally open(NO) Contact	
Capacitive sensor		Normally Close(NC) Contact	
Optical sensor		Lamp	
Ultrasonic sensor		Multimeter	

Optical Sensor Symbols

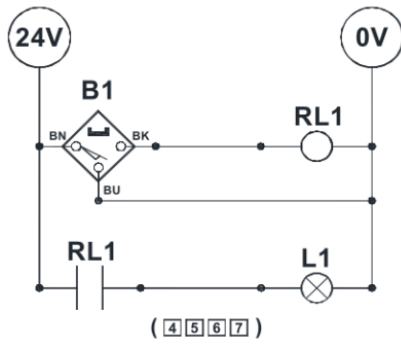
Description	Symbol	Description	Symbol
Diffuse beam sensor		Diffuse beam with background suppression sensor	
Thru beam sensor		Retro reflective sensor	

List of Experiments

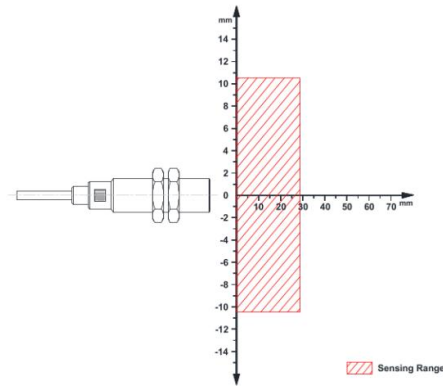
1. Operation of Proximity Sensors - Magnetic field sensor, Inductive sensor - flush head & non-flush head and Capacitive sensor - flush head
2. Operation of Optical Sensors – Diffuse beam and Diffuse beam with background suppression, Thru beam and retro reflective using reflector
3. Operation of Fiber optic sensors – Diffuse beam and Thru beam
4. Operation of Colour Sensor
5. Operation of Ultrasonic Sensor
6. Execution of digital sensors with PLC
7. Execution of analog sensor with PLC using I/O link devices
8. Execution of digital sensor with PLC using I/O link devices

Magnetic Sensor:

Relay Ladder Diagram

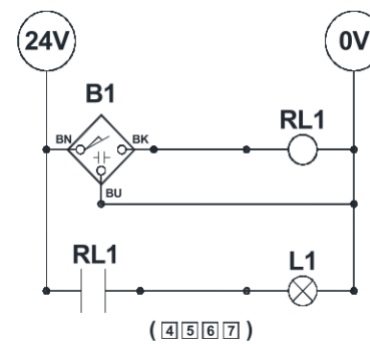


Model Plot

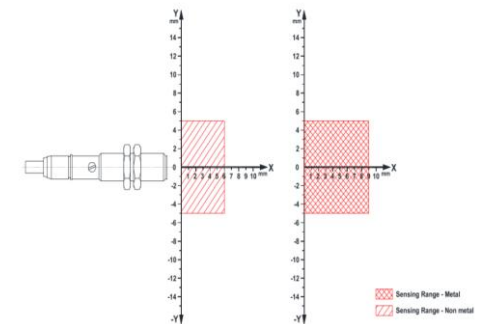


Capacitive Sensor:

Relay Ladder Diagram

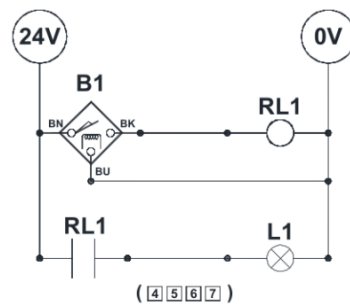


Model Plot

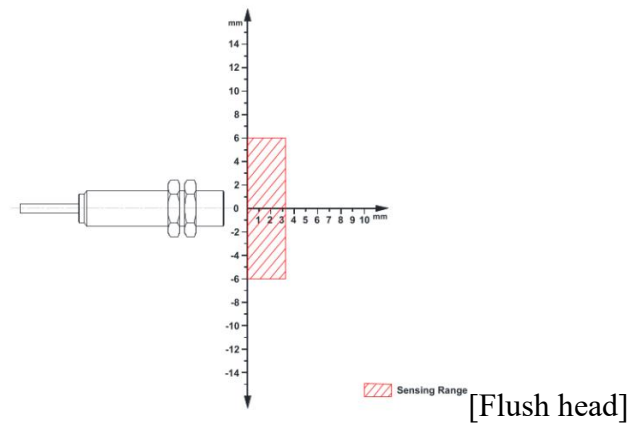


Inductive Sensor:

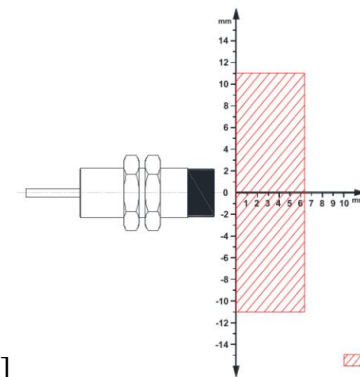
Relay Ladder Diagram



Model Plot



[Flush head]



[Non-Flush head]

Ex. No.	Operation of Proximity Sensors
Date:	

Aim:

To observe the operation of Proximity Sensors

Part List:

S. No.	Product Name	Product No.	Quantity
1	Magnetic field sensor- Flush head	MP0223	1
2	Inductive sensor - Flush head	MP0224	1
3	Inductive sensor- non-flush head	MP0225	1
4	Capacitive sensor - Flush head	MP0227	1
5	LM guide unit	MP0239	1
6	Workpiece for magnetic, inductive & capacitive sensor	MP0259-01, MP0259-02	1
7	Electrical Push button module	MP0059	1
8	Electrical connection set	MP0027	1
9	Relay logic unit 4C/O - 3 Relays	MP0074	1
10	Switch mode power supply	JTK-SMPS	1

Test Procedure:

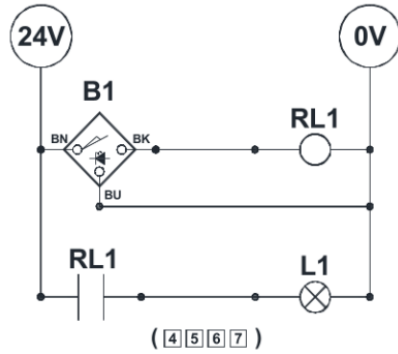
1. Arrange the components and make the electrical connection as per the relay ladder diagram.
2. Locate the workpiece and sensor head in the same axis and set the observed value in the horizontal scale as 0.
3. With the vertical scale value as 0, move the workpiece left until the output of the sensor cutoff. Mark the value and plot it in observation plot. Repeat the step in the opposite direction and plot the value in the observation.
4. Similarly make the observations by increasing the values in vertical scale.

Inference:

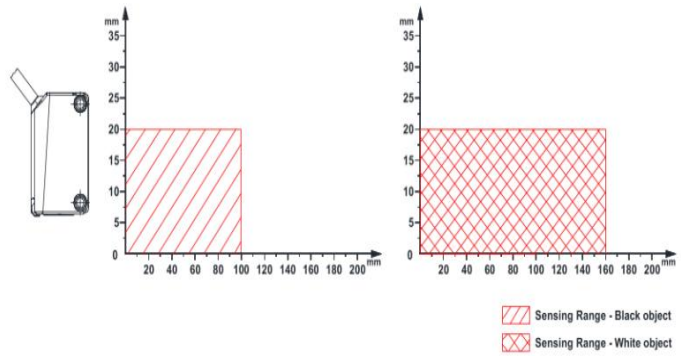
Result:

Diffuse Beam:

Relay Ladder Diagram

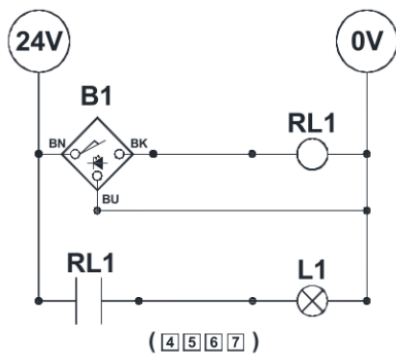


Model Plot

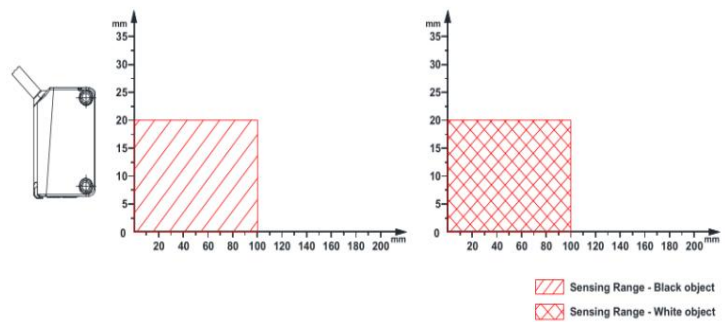


Diffuse beam with background suppression:

Relay Ladder Diagram



Model Plot



Ex. No.	Operation of Optical Sensors
Date:	

Aim:

To observe the operation of Optical Sensors - Diffuse beam and Diffuse beam with background suppression, Thru beam and retro reflective using reflector

Part List:

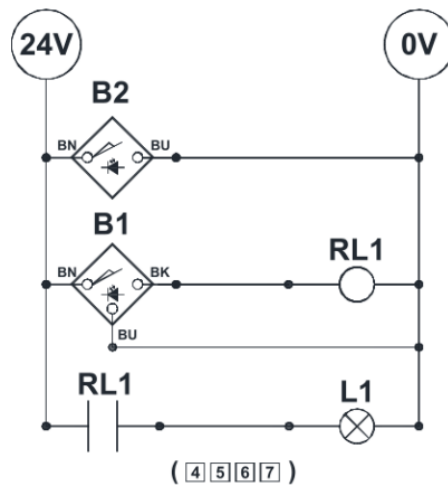
S. No.	Product Name	Product No.	Quantity
1	Optical sensor - Diffuse beam	MP0233	1
2	Optical sensor - Diffuse beam with background suppression	MP0232	1
3	Optical sensor - Thru beam	MP0228	1
4	Optical sensor – Retro Reflective	MP0229	1
5	LM guide unit	MP0239	1
6	Workpiece for optical sensor	MP0259-03	1
7	Electrical Push button module	MP0059	1
8	Electrical connection set	MP0027	1
9	Relay logic unit 4C/O - 3 Relays	MP0074	1
10	Switch mode power supply	JTK-SMPS	1

Test Procedure:

1. Arrange the components and make the electrical connection as per the relay ladder diagram.
2. Set the black object in the workpiece and adjust vertically until the light beam from the sensor touches the workpiece. Observe the value and move the workpiece in horizontal direction until the output is cutoff and plot the values in observation plot
3. Similarly make the observations by switching white object and plot the values of observation as well.
4. For Thru beam & retro reflective, arrange the components and make the electrical connection as per the relay ladder diagram.
5. Move the workpiece in-between the emitter (B1) and receiver (B2) and observe that output will cutoff. – Thru Beam
6. Move the workpiece in-between the emitter (B1) and reflector and observe that output will cutoff. – Retro Reflective

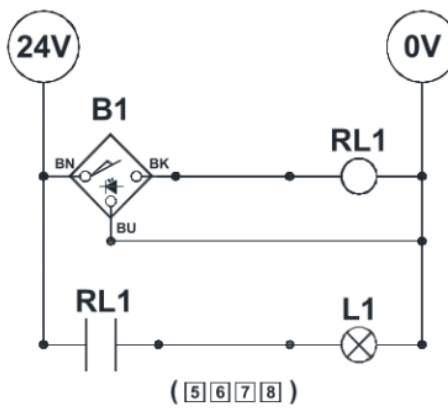
Thru Beam:

Relay Ladder Diagram



Retro reflective using reflector:

Relay Ladder Diagram

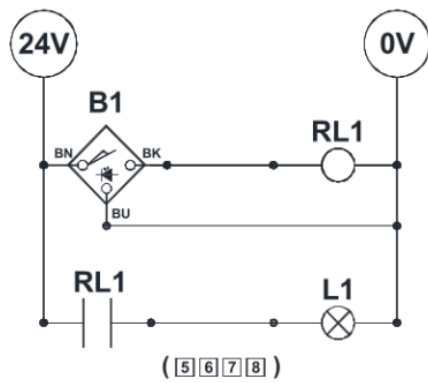


Inference:

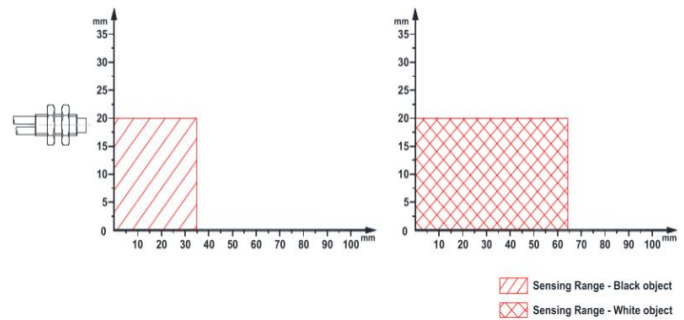
Result:

Fiber Optic Sensor – Diffuse beam:

Relay Ladder Diagram

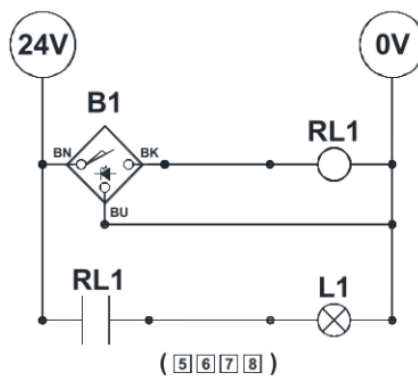


Model Plot



Fiber Optic Sensor – Thru beam:

Relay Ladder Diagram



Ex. No.	Operation of Fiber optic sensors – Diffuse beam and Thru beam
Date:	

Aim:

To observe the operation of Fiber Optical Sensors - Diffuse beam and Thru beam

Part List:

S. No.	Product Name	Product No.	Quantity
1	Fiber optic sensor amplifier unit	MP0234	1
2	Fiber optic sensor - Diffuse beam	MP0235	1
3	Fiber optic sensor - Thru beam	MP0236	1
4	LM guide unit	MP0239	1
5	Workpiece for optical sensor	MP0259-03	1
6	Electrical Push button module	MP0059	1
7	Electrical connection set	MP0027	1
8	Relay logic unit 4C/O - 3 Relays	MP0074	1
9	Switch mode power supply	JTK-SMPS	1

Test Procedure:

Fiber optic sensor - Diffuse beam

1. Arrange the components and make the electrical connection as per the relay ladder diagram.
2. Set the black object in the workpiece at 0 in horizontal scale. Move the sensor close to the object until the reading in the amplifier unit becomes 0.6. Increase the value in horizontal scale until the output is present plot the horizontal scale value in observation plot.
3. Repeat the above step for white object in the workpiece and plot the horizontal scale value in observation as well.

Fiber optic sensor - Thru beam

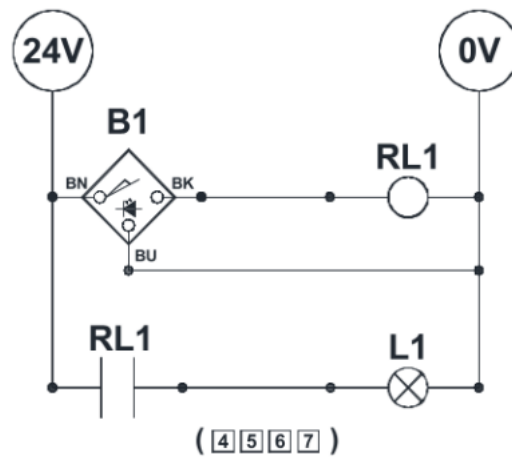
1. Arrange the components and make the electrical connection as per the relay ladder diagram.
2. Move the workpiece in-between the emitter and receiver and observe that output will cutoff.

Inference:

Result:

Colour Sensor:

Relay Ladder Diagram



Sensing Range:

Actual Sensing Range:

Samples					
Sample no.	1	2	3	4	5
Status	✓	✓	✓	✗	✗

✓ - Detected
✗ - Not detected

Observation:

Samples					
Sample no.	1	2	3	4	5
Status					

Ex. No.	Operation of Colour Sensor
Date:	

Aim:

To observe the operation of Colour Sensor

Part List:

S. No.	Product Name	Product No.	Quantity
1	Colour sensor	MP0237	1
2	LM guide unit	MP0239	1
3	Workpiece for Colour sensor	MP0259-04	1
4	Electrical Push button module	MP0059	1
5	Electrical connection set	MP0027	1
6	Relay logic unit 4C/O - 3 Relays	MP0074	1
7	Switch mode power supply	JTK-SMPS	1

Test Procedure:

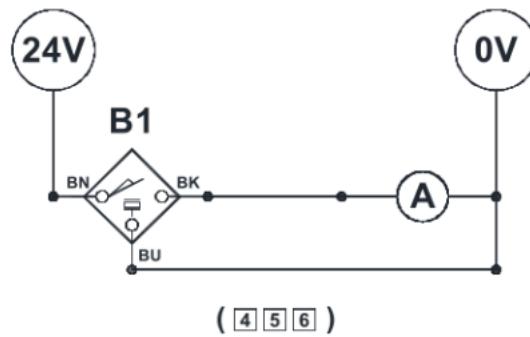
1. Arrange the components and make the electrical connection as per the relay ladder diagram.
2. Set the black colour (sample no.1) in the workpiece and mark the output status in the observation on respective column. Similarly repeat the steps for other sample colours as well.

Inference:

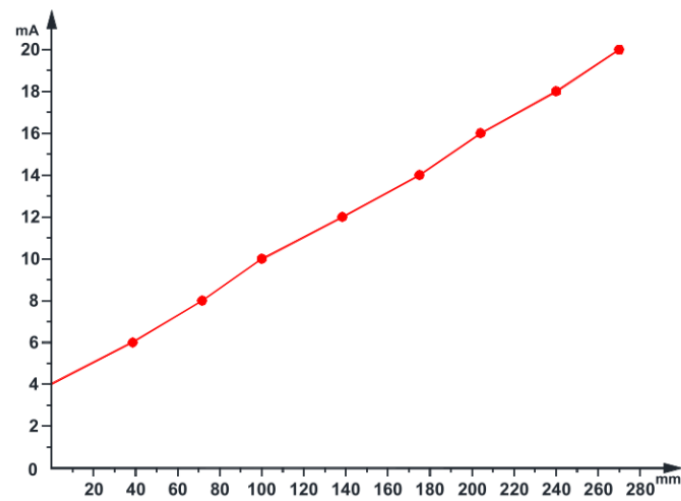
Result:

Ultrasonic Sensor:

Relay Ladder Diagram



Model Plot



Observation Table

Ex. No.	Operation of Ultrasonic Sensor
Date:	

Aim:

To observe the operation of Ultrasonic Sensor

Part List:

S. No.	Product Name	Product No.	Quantity
1	Ultrasonic sensor	MP0238	1
2	LM guide unit	MP0239	1
3	Workpiece for Ultrasonic sensor	MP0259-03	1
4	Electrical Push button module	MP0059	1
5	Electrical connection set	MP0027	1
6	Relay logic unit 4C/O - 3 Relays	MP0074	1
7	Switch mode power supply	JTK-SMPS	1

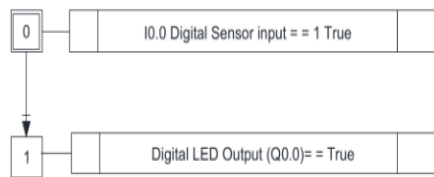
Test Procedure:

1. Arrange the components and make the electrical connection as per the relay ladder diagram
2. Align the workpiece and sensor for the lower evaluation limit and set the observed reading in the horizontal scale as 0. Mark the mA value in the multimeter and plot it in observation.
3. Similarly make the observations by increasing the values in horizontal scale and observe the mA value in the multimeter.

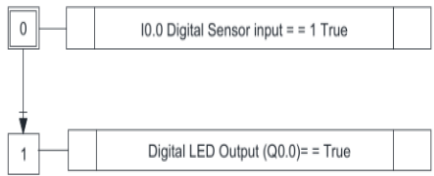
Inference:

Result:

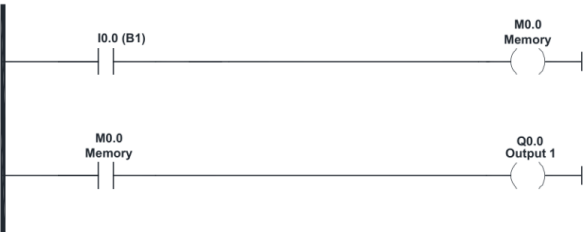
**Magnetic Sensor:
Sequential Flowchart:**



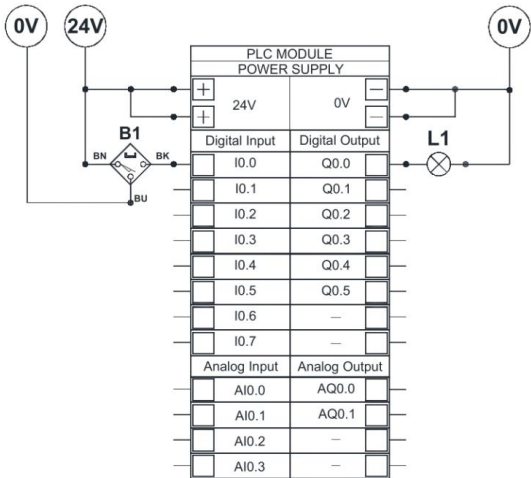
**Optical Sensor:
Sequential Flowchart:**



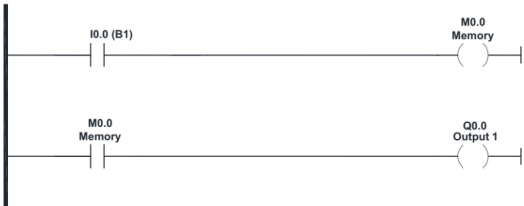
PLC Program:



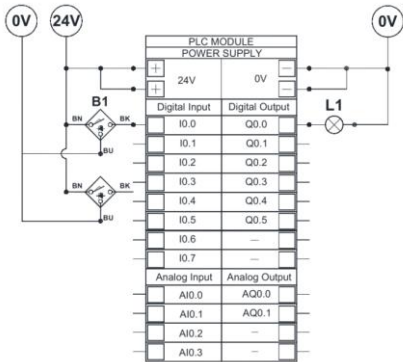
Schematic Diagram:



PLC Program:



Schematic Diagram:



Ex. No.	Execution of digital sensor with PLC
Date:	

Aim:

To execute and observe the operation of digital sensors using PLC

Part List:

S. No.	Product Name	Product No.	Quantity
1	Magnetic field sensor- Flush head	MP0223	1
2	Optical sensor - Thru beam	MP0228	1
3	LM guide unit	MP0239	1
4	Workpiece for magnetic, inductive & capacitive sensor	MP0259-01, MP0259-02	1
5	Electrical Push button module	MP0059	1
6	Electrical connection set	MP0027	1
7	Relay logic unit 4C/O - 3 Relays	MP0074	1
8	Switch mode power supply	JTK-SMPS	1

Test Procedure:

Magnetic Sensor:

1. Arrange the components and make the electrical connection as per the relay ladder diagram.
2. Locate the workpiece and sensor head in the same axis and set the observed value in the horizontal scale as 0.
3. With the vertical scale value as 0, move the workpiece left until the output of the sensor cutoff. Mark the value and plot it in observation plot. Repeat the step in the opposite direction and plot the value in the observation.
4. Similarly make the observations by increasing the values in vertical scale.

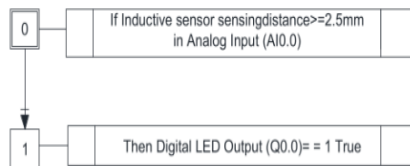
Optical Sensor:

1. Arrange the components and make the electrical connection as per the relay ladder diagram.
2. Move the workpiece in-between the emitter (B1) and receiver (B2) and observe that output will cutoff. – Thru Beam

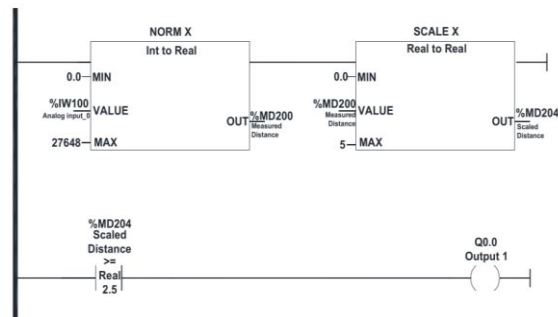
Inference:

Result:

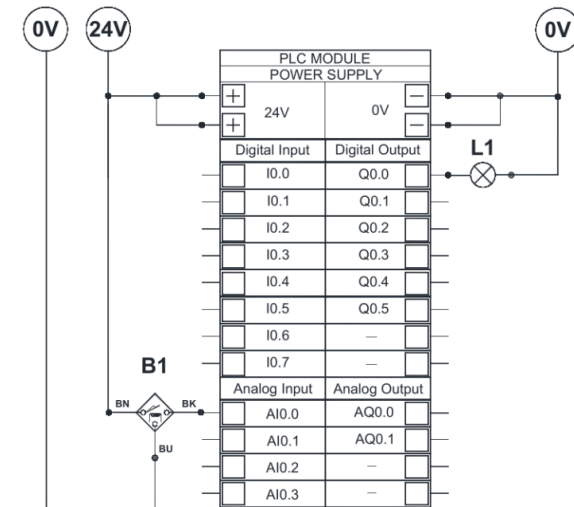
Inductive Analog Sensor – Flush Type: Sequential Flowchart:



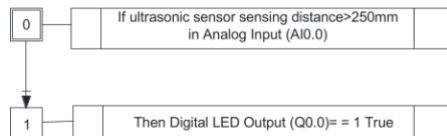
PLC Program:



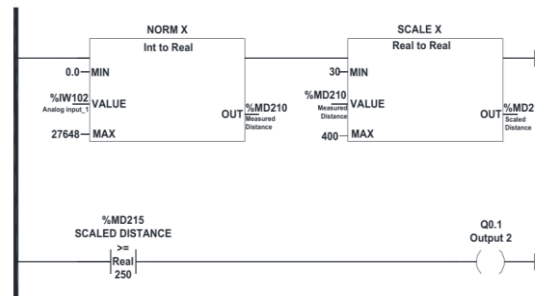
Schematic Diagram:



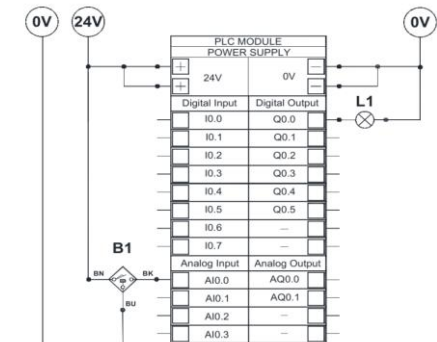
Ultrasonic Sensor: Sequential Flowchart:



PLC Program:



Schematic Diagram:



Ex. No.	Execution of analog sensor with PLC using I/O link devices
Date:	

Aim:

To execute and observe the operation of analog sensors using PLC with I/O module

Part List:

S. No.	Product Name	Product No.	Quantity
1	Inductive analog sensor- Flush head	MP0226	1
2	Ultrasonic Sensor	MP0238	1
3	Turck Analog Module		1
4	Workpiece for inductive & ultrasonic sensor	MP0259-01	1
5	Electrical Push button module	MP0059	1
6	Electrical connection set	MP0027	1
7	Relay logic unit 4C/O - 3 Relays	MP0074	1
8	Switch mode power supply	JTK-SMPS	1
9	Ethernet Cables		3

Test Procedure:

Inductive analog Sensor:

1. Arrange the components as per the relay diagram
2. Configure the turck module and interface it with PLC
3. Write the PLC program and validate the results
4. Similarly observe the readings for ultrasonic sensor

Inference:

Result:

Magnetic Coded Switch:

PLC Program:

Schematic Diagram:

Ex. No.	Execution of digital sensor with PLC using I/O link devices
Date:	

Aim:

To execute and observe the operation of digital sensors using PLC with I/O module.

Part List:

S. No.	Product Name	Product No.	Quantity
1	Digital Sensor – Magnetic Coded Switch		1
2	Turck Digital Module		1
3	Ethernet Cables	-	3
4	Electrical Push button module	MP0059	1
5	Electrical connection set	MP0027	1
6	Relay logic unit 4C/O - 3 Relays	MP0074	1
7	Switch mode power supply	JTK-SMPS	1

Test Procedure:

Magnetic Coded Switch:

5. Arrange the components as per the relay diagram
6. Configure the turck module and interface it with PLC
7. Write the PLC program and validate the results

Inference:

Result:

